Introduction

Algorithms to Live By

Imagine you're searching for an apartment in San Francisco—arguably the most harrowing American city in which to do so. The booming tech sector and tight zoning laws limiting new construction have conspired to make the city just as expensive as New York, and by many accounts more competitive. New listings go up and come down within minutes, open houses are mobbed, and often the keys end up in the hands of whoever can physically foist a deposit check on the landlord first.

Such a savage market leaves little room for the kind of fact-finding and deliberation that is theoretically supposed to characterize the doings of the rational consumer. Unlike, say, a mall patron or an online shopper, who can compare options before making a decision, the would-be San Franciscan has to decide instantly either way: you can take the apartment you are currently looking at, forsaking all others, or you can walk away, never to return.

Let's assume for a moment, for the sake of simplicity, that you care only about maximizing your chance of getting the very best apartment available. Your goal is reducing the twin, Scylla-and-Charybdis regrets of the "one that got away" and the "stone left unturned" to the absolute minimum. You run into a dilemma right off the bat: How are you to know that an apartment is indeed the best unless you have a baseline to judge it by? And how are you to establish that baseline unless you look at (and *lose*) a number of apartments? The more information you gather, the better you'll know the right opportunity when you see it—but the more likely you are to have already passed it by.

So what do you do? How do you make an informed decision when the

very act of informing it jeopardizes the outcome? It's a cruel situation, bordering on paradox.

When presented with this kind of problem, most people will intuitively say something to the effect that it requires some sort of balance between looking and leaping—that you must look at enough apartments to establish a standard, then take whatever satisfies the standard you've established. This notion of balance is, in fact, precisely correct. What most people *don't* say with any certainty is what that balance is. Fortunately, there's an answer.

Thirty-seven percent.

If you want the best odds of getting the best apartment, spend 37% of your apartment hunt (eleven days, if you've given yourself a month for the search) noncommittally exploring options. Leave the checkbook at home; you're just calibrating. But after that point, be prepared to immediately commit—deposit and all—to the very first place you see that beats whatever you've already seen. This is not merely an intuitively satisfying compromise between looking and leaping. It is the *provably optimal* solution.

We know this because finding an apartment belongs to a class of mathematical problems known as "optimal stopping" problems. The 37% rule defines a simple series of steps—what computer scientists call an "algorithm"—for solving these problems. And as it turns out, apartment hunting is just one of the ways that optimal stopping rears its head in daily life. Committing to or forgoing a succession of options is a structure that appears in life again and again, in slightly different incarnations. How many times to circle the block before pulling into a parking space? How far to push your luck with a risky business venture before cashing out? How long to hold out for a better offer on that house or car?

The same challenge also appears in an even more fraught setting: dating. Optimal stopping is the science of serial monogamy.

Simple algorithms offer solutions not only to an apartment hunt but to all such situations in life where we confront the question of optimal stopping. People grapple with these issues every day—although surely poets have spilled more ink on the tribulations of courtship than of parking—and they do so with, in some cases, considerable anguish. But the anguish is unnecessary. Mathematically, at least, these are solved

1 Optimal Stopping

When to Stop Looking

> *Though all Christians start a wedding invitation by solemnly declaring their marriage is due to special Divine arrangement, I, as a philosopher, would like to talk in greater detail about this…*
>
> —JOHANNES KEPLER

> *If you prefer Mr. Martin to every other person; if you think him the most agreeable man you have ever been in company with, why should you hesitate?*
>
> —JANE AUSTEN, *EMMA*

It's such a common phenomenon that college guidance counselors even have a slang term for it: the "turkey drop." High-school sweethearts come home for Thanksgiving of their freshman year of college and, four days later, return to campus single.

An angst-ridden Brian went to his own college guidance counselor his freshman year. His high-school girlfriend had gone to a different college several states away, and they struggled with the distance. They also struggled with a stranger and more philosophical question: how good a relationship did they have? They had no real benchmark of other relationships by which to judge it. Brian's counselor recognized theirs as a classic freshman-year dilemma, and was surprisingly nonchalant in her advice: "Gather data."

The nature of serial monogamy, writ large, is that its practitioners are confronted with a fundamental, unavoidable problem. When have you met enough people to know who your best match is? And what if acquiring the data costs you that very match? It seems the ultimate

Catch-22 of the heart.

As we have seen, this Catch-22, this angsty freshman cri de coeur, is what mathematicians call an "optimal stopping" problem, and it may actually have an answer: 37%.

Of course, it all depends on the assumptions you're willing to make about love.

The Secretary Problem

In any optimal stopping problem, the crucial dilemma is not which option to *pick,* but how many options to even *consider*. These problems turn out to have implications not only for lovers and renters, but also for drivers, homeowners, burglars, and beyond.

The **37% Rule**[*1] derives from optimal stopping's most famous puzzle, which has come to be known as the "secretary problem." Its setup is much like the apartment hunter's dilemma that we considered earlier. Imagine you're interviewing a set of applicants for a position as a secretary, and your goal is to maximize the chance of hiring the single best applicant in the pool. While you have no idea how to assign scores to individual applicants, you can easily judge which one you prefer. (A mathematician might say you have access only to the *ordinal* numbers—the relative ranks of the applicants compared to each other—but not to the *cardinal* numbers, their ratings on some kind of general scale.) You interview the applicants in random order, one at a time. You can decide to offer the job to an applicant at any point and they are guaranteed to accept, terminating the search. But if you pass over an applicant, deciding not to hire them, they are gone forever.

The secretary problem is widely considered to have made its first appearance in print—sans explicit mention of secretaries—in the February 1960 issue of *Scientific American,* as one of several puzzles posed in Martin Gardner's beloved column on recreational mathematics. But the origins of the problem are surprisingly mysterious. Our own initial search yielded little but speculation, before turning into unexpectedly physical detective work: a road trip down to the archive of Gardner's papers at Stanford, to haul out boxes of his midcentury correspondence. Reading paper correspondence is a bit like

eavesdropping on someone who's on the phone: you're only hearing one side of the exchange, and must infer the other. In our case, we only had the replies to what was apparently Gardner's own search for the problem's origins fiftysome years ago. The more we read, the more tangled and unclear the story became.

Harvard mathematician Frederick Mosteller recalled hearing about the problem in 1955 from his colleague Andrew Gleason, who had heard about it from somebody else. Leo Moser wrote from the University of Alberta to say that he read about the problem in "some notes" by R. E. Gaskell of Boeing, who himself credited a colleague. Roger Pinkham of Rutgers wrote that he first heard of the problem in 1955 from Duke University mathematician J. Shoenfield, "and I believe he said that he had heard the problem from someone at Michigan."

"Someone at Michigan" was almost certainly someone named Merrill Flood. Though he is largely unheard of outside mathematics, Flood's influence on computer science is almost impossible to avoid. He's credited with popularizing the traveling salesman problem (which we discuss in more detail in chapter 8), devising the prisoner's dilemma (which we discuss in chapter 11), and even with possibly coining the term "software." It's Flood who made the first known discovery of the 37% Rule, in 1958, and he claims to have been considering the problem since 1949—but he himself points back to several other mathematicians.

Suffice it to say that wherever it came from, the secretary problem proved to be a near-perfect mathematical puzzle: simple to explain, devilish to solve, succinct in its answer, and intriguing in its implications. As a result, it moved like wildfire through the mathematical circles of the 1950s, spreading by word of mouth, and thanks to Gardner's column in 1960 came to grip the imagination of the public at large. By the 1980s the problem and its variations had produced so much analysis that it had come to be discussed in papers as a subfield unto itself.

As for secretaries—it's charming to watch each culture put its own anthropological spin on formal systems. We think of chess, for instance, as medieval European in its imagery, but in fact its origins are in eighth-century India; it was heavy-handedly "Europeanized" in the fifteenth century, as its shahs became kings, its viziers turned to queens, and its elephants became bishops. Likewise, optimal stopping problems have

had a number of incarnations, each reflecting the predominating concerns of its time. In the nineteenth century such problems were typified by baroque lotteries and by women choosing male suitors; in the early twentieth century by holidaying motorists searching for hotels and by male suitors choosing women; and in the paper-pushing, male-dominated mid-twentieth century, by male bosses choosing female assistants. The first explicit mention of it by name as the "secretary problem" appears to be in a 1964 paper, and somewhere along the way the name stuck.

Whence 37%?

In your search for a secretary, there are two ways you can fail: stopping early and stopping late. When you stop too early, you leave the best applicant undiscovered. When you stop too late, you hold out for a better applicant who doesn't exist. The optimal strategy will clearly require finding the right balance between the two, walking the tightrope between looking too much and not enough.

If your aim is finding the very best applicant, settling for nothing less, it's clear that as you go through the interview process you shouldn't even consider hiring somebody who isn't the best you've seen so far. However, simply being the best yet isn't enough for an offer; the very first applicant, for example, will of course be the best yet by definition. More generally, it stands to reason that the rate at which we encounter "best yet" applicants will go down as we proceed in our interviews. For instance, the second applicant has a 50/50 chance of being the best we've yet seen, but the fifth applicant only has a 1-in-5 chance of being the best so far, the sixth has a 1-in-6 chance, and so on. As a result, best-yet applicants will become steadily more impressive as the search continues (by definition, again, they're better than all those who came before)—but they will also become more and more infrequent.

Okay, so we know that taking the *first* best-yet applicant we encounter (a.k.a. the first applicant, period) is rash. If there are a hundred applicants, it also seems hasty to make an offer to the *next* one who's best-yet, just because she was better than the first. So how do we proceed?

Intuitively, there are a few potential strategies. For instance, making an offer the third time an applicant trumps everyone seen so far—or maybe the fourth time. Or perhaps taking the next best-yet applicant to come along after a long "drought"—a long streak of poor ones.

But as it happens, neither of these relatively sensible strategies comes out on top. Instead, the optimal solution takes the form of what we'll call the **Look-Then-Leap Rule**: You set a predetermined amount of time for "looking"—that is, exploring your options, gathering data—in which you categorically don't choose anyone, no matter how impressive. After that point, you enter the "leap" phase, prepared to instantly commit to anyone who outshines the best applicant you saw in the look phase.

We can see how the Look-Then-Leap Rule emerges by considering how the secretary problem plays out in the smallest applicant pools. With just one applicant the problem is easy to solve—hire her! With two applicants, you have a 50/50 chance of success no matter what you do. You can hire the first applicant (who'll turn out to be the best half the time), or dismiss the first and by default hire the second (who is also best half the time).

Add a third applicant, and all of a sudden things get interesting. The odds if we hire at random are one-third, or 33%. With two applicants we could do no better than chance; with three, can we? It turns out we can, and it all comes down to what we do with the second interviewee. When we see the first applicant, we have no *information*—she'll always appear to be the best yet. When we see the third applicant, we have no *agency*—we have to make an offer to the final applicant, since we've dismissed the others. But when we see the second applicant, we have a little bit of both: we know whether she's better or worse than the first, and we have the freedom to either hire or dismiss her. What happens when we just hire her if she's better than the first applicant, and dismiss her if she's not? This turns out to be the best possible strategy when facing three applicants; using this approach it's possible, surprisingly, to do just as well in the three-applicant problem as with two, choosing the best applicant exactly half the time.[*2]

Enumerating these scenarios for four applicants tells us that we should still begin to leap as soon as the second applicant; with five applicants in the pool, we shouldn't leap before the third.

As the applicant pool grows, the exact place to draw the line between

looking and leaping settles to 37% of the pool, yielding the 37% Rule: look at the first 37% of the applicants,[*3] choosing none, then be ready to leap for anyone better than all those you've seen so far.

Number of Applicants	Take the Best Applicant After	Chance of Getting the Best
3	1 (33.33%)	50%
4	1 (25%)	45.83%
5	2 (40%)	43.33%
6	2 (33.33%)	42.78%
7	2 (28.57%)	41.43%
8	3 (37.5%)	40.98%
9	3 (33.33%)	40.59%
10	3 (30%)	39.87%
20	7 (35%)	38.42%
30	11 (36.67%)	37.86%
40	15 (37.5%)	37.57%
50	18 (36%)	37.43%
100	37 (37%)	37.10%
1000	369 (36.9%)	36.81%

How to optimally choose a secretary.

As it turns out, following this optimal strategy ultimately gives us a 37% chance of hiring the best applicant; it's one of the problem's curious mathematical symmetries that the strategy itself and its chance of success work out to the very same number. The table above shows the optimal strategy for the secretary problem with different numbers of applicants, demonstrating how the chance of success—like the point to switch from looking to leaping—converges on 37% as the number of applicants increases.

A 63% failure rate, when following the *best possible* strategy, is a sobering fact. Even when we act optimally in the secretary problem, we will still fail most of the time—that is, we won't end up with the single best applicant in the pool. This is bad news for those of us who would

frame romance as a search for "the one." But here's the silver lining. Intuition would suggest that our chances of picking the single best applicant should steadily decrease as the applicant pool grows. If we were hiring at random, for instance, then in a pool of a hundred applicants we'd have a 1% chance of success, and in a pool of a million applicants we'd have a 0.0001% chance. Yet remarkably, the math of the secretary problem doesn't change. If you're stopping optimally, your chance of finding the single best applicant in a pool of a hundred is 37%. And in a pool of a million, believe it or not, your chance is still 37%. Thus the bigger the applicant pool gets, the more valuable knowing the optimal algorithm becomes. It's true that you're unlikely to find the needle the majority of the time, but optimal stopping is your best defense against the haystack, no matter how large.

Lover's Leap

> *The passion between the sexes has appeared in every age to be so nearly the same that it may always be considered, in algebraic language, as a given quantity.*
>
> —THOMAS MALTHUS

> *I married the first man I ever kissed. When I tell this to my children they just about throw up.*
>
> —BARBARA BUSH

Before he became a professor of operations research at Carnegie Mellon, Michael Trick was a graduate student, looking for love. "It hit me that the problem has been studied: it is the Secretary Problem! I had a position to fill [and] a series of applicants, and my goal was to pick the best applicant for the position." So he ran the numbers. He didn't know how many women he could expect to meet in his lifetime, but there's a certain flexibility in the 37% Rule: it can be applied to either the number of applicants or the *time* over which one is searching. Assuming that his search would run from ages eighteen to forty, the 37% Rule gave age 26.1 years as the point at which to switch from looking to leaping. A number that, as it happened, was exactly Trick's age at the time. So when he found a woman who was a better match than all those he had dated so